

ANNAPAREDDY PAVAN KUMAR REDDY

DevOps Engineer | Site Reliability Engineer | AWS & Kubernetes

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PROFESSIONAL SUMMARY

DevOps/SRE Engineer with 4+ years of IT experience, including 3+ years building CI/CD pipelines, provisioning cloud infrastructure as code, and operating Kubernetes workloads on AWS. Hands-on expertise spans Terraform-provisioned EKS clusters, GitOps delivery with ArgoCD, Horizontal Pod Autoscaling validated under real load, and CI/CD pipelines using Jenkins and GitHub Actions. Skilled at diagnosing production-grade infrastructure issues — from EKS scheduling constraints to TLS certificate failures — and resolving them end to end.

TECHNICAL SKILLS

- **Cloud:** AWS (EC2, S3, IAM, VPC, EKS, ECR, CloudWatch, ELB)
- **IaC:** Terraform (modular IaC, remote state, IAM roles)
- **Containers & Orchestration:** Docker, Docker Compose, Kubernetes (EKS), Helm
- **Scaling & GitOps:** HPA (Horizontal Pod Autoscaler), Metrics Server, ArgoCD (GitOps, Self-Heal)
- **CI/CD:** Jenkins, GitHub Actions
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack, CloudWatch
- **Configuration Management:** Ansible (playbooks, roles, handlers, variables)
- **Version Control:** Git, GitHub, GitOps workflows
- **Linux & Scripting:** Ubuntu, CentOS, Amazon Linux, Bash scripting
- **Programming:** Python (Flask), MySQL, YAML, JSON, Groovy
- **Security:** RBAC, IAM least-privilege policies, Security Groups, Kubernetes Secrets

PROFESSIONAL EXPERIENCE

DevOps Engineer | Tech Mahindra

Oct 2021 – Feb 2025

- Designed and maintained CI/CD pipelines using Jenkins and GitHub Actions, reducing deployment time by 40% and enabling automated build, test, and deployment workflows across multiple environments.
- Provisioned and managed AWS infrastructure (EC2, VPC, S3, IAM, EKS) using Terraform, achieving consistent and reproducible environment setups across dev, staging, and production.
- Containerized microservices using Docker and orchestrated deployments on Kubernetes (EKS), managing Pods, Deployments, Services, ConfigMaps, Secrets, Ingress, and HPA.
- Implemented GitOps-based continuous delivery using ArgoCD, enabling automatic synchronization of Kubernetes cluster state with GitHub repository configurations.
- Packaged and deployed applications using Helm charts, streamlining multi-environment deployments and enabling one-command rollbacks.
- Set up monitoring and alerting using Prometheus and Grafana, achieving 99.9% uptime visibility across services; integrated ELK Stack for centralized log management.
- Configured RBAC policies in Kubernetes to enforce least-privilege access for development teams across namespaces.
- Received Bravo Award for outstanding contribution to infrastructure automation and deployment reliability improvements.

IT Support Engineer | Megna Super IT Solutions

Jun 2020 – Oct 2021

- Provided L1/L2 technical support for Linux and Windows systems, resolving 95%+ of tickets within SLA timelines.
- Automated routine system administration tasks using Bash scripts, reducing manual effort by 30%.
- Managed user accounts, permissions, and system configurations across on-premise infrastructure.
- Monitored system health and proactively resolved incidents to minimize downtime.

CI/CD & GITOPS

- Built multi-stage GitHub Actions pipelines (test → build → deploy) with job dependencies (needs:), tagging Docker images with unique Git commit SHAs instead of mutable tags to guarantee reproducible deployments.
- Automated Kubernetes manifest updates within CI — pipeline rewrites image tags in Deployment YAML and commits back to Git, triggering downstream GitOps sync rather than deploying directly from CI.
- Configured ArgoCD with Automatic sync and Self-Heal policies; validated under chaos testing by manually deleting live production pods and confirming full state restoration within seconds, with zero manual kubectl intervention.
- Designed CI/CD on Jenkins using Declarative Pipelines and Jenkinsfile-as-code, including multi-stage builds (Checkout, Build, Test, Deploy) integrated with Docker.
- Secured pipeline credentials using GitHub Actions Secrets and Docker Hub access tokens, avoiding hardcoded credentials in any workflow file.
- Used in: Self-Healing GitOps Platform, 3-Tier Blog Platform, Flask DevOps App — github.com/Pavan-7673/blog-platform | github.com/Pavan-7673/devops-app-flask

INFRASTRUCTURE AS CODE (TERRAFORM)

- Provisioned a production-grade AWS EKS cluster from scratch using modular Terraform split across 7 files (provider.tf, variables.tf, data.tf, iam.tf, eks.tf, outputs.tf, versions.tf), following IaC separation-of-concerns best practices.
- Created dedicated least-privilege IAM roles for the EKS control plane and worker nodes (AmazonEKSClusterPolicy, AmazonEKS_CNI_Policy, AmazonEC2ContainerRegistryReadOnly), avoiding over-permissioned service roles.
- Used Terraform data sources to reference existing AWS VPC/subnets instead of duplicating networking resources, reducing infrastructure footprint and cost.
- Managed full infrastructure lifecycle with terraform plan/apply/destroy, enabling clean teardown of non-production environments to control AWS spend.
- Diagnosed and resolved an EKS ENI-based pod-per-node scheduling limit on t3.small worker nodes during a scale-out event, right-sizing the node group via AWS CLI to restore cluster scheduling capacity.
- Used in: Ecommerce App on AWS EKS, Self-Healing GitOps Platform — github.com/Pavan-7673/terraform-eks | github.com/Pavan-7673/ecommerce-devops-app

KUBERNETES & CONTAINER ORCHESTRATION

- Deployed and operated 3-tier production workloads on AWS EKS (nginx frontend, Flask REST API backend, MySQL 8.0 database), each in dedicated Deployments within isolated namespaces.
- Implemented Horizontal Pod Autoscaling (HPA) on frontend and backend tiers; load-tested with synthetic traffic and observed live scale-out from 2 to 6 backend pods as CPU utilization reached 99%, fully managed via GitOps rather than imperative kubectl commands.
- Diagnosed and fixed a Metrics Server TLS certificate verification failure that was silently blocking all HPA metrics collection cluster-wide, restoring live CPU/memory telemetry via a targeted kubelet configuration change.
- Secured workloads using Kubernetes Secrets for credential injection (eliminating hardcoded database passwords), RBAC (Role, RoleBinding, ServiceAccount) for least-privilege namespace access, and resource requests/limits to prevent resource starvation across tiers.
- Exposed services using ClusterIP (internal tier-to-tier communication) and LoadBalancer (external access via AWS Elastic Load Balancer with public DNS), and packaged applications as Helm charts with parameterized values.yaml for dev/staging/prod deployments without code changes.
- Used in: Self-Healing GitOps Platform, 3-Tier Blog Platform, Flask DevOps App — github.com/Pavan-7673/blog-platform | github.com/Pavan-7673/devops-app-flask

EDUCATION

B.Tech in Computer Science Engineering

Acharya Nagarjuna University

CERTIFICATIONS & ACHIEVEMENTS

- Bravo Award — Tech Mahindra (2023) — Recognized for outstanding contribution to infrastructure automation
- Pat on the Back Award — Tech Mahindra — Recognized for proactive problem-solving and reliability contributions